

In a shipyard, a loaded transporter does not drive on the road. It owns the road.

Network-bound capacity, deadlock-free orchestration and fleet-network co-design
(working title)

Road-occupying transport in shipyards: network-bound capacity, deadlock-free orchestration and fleet-network co-design

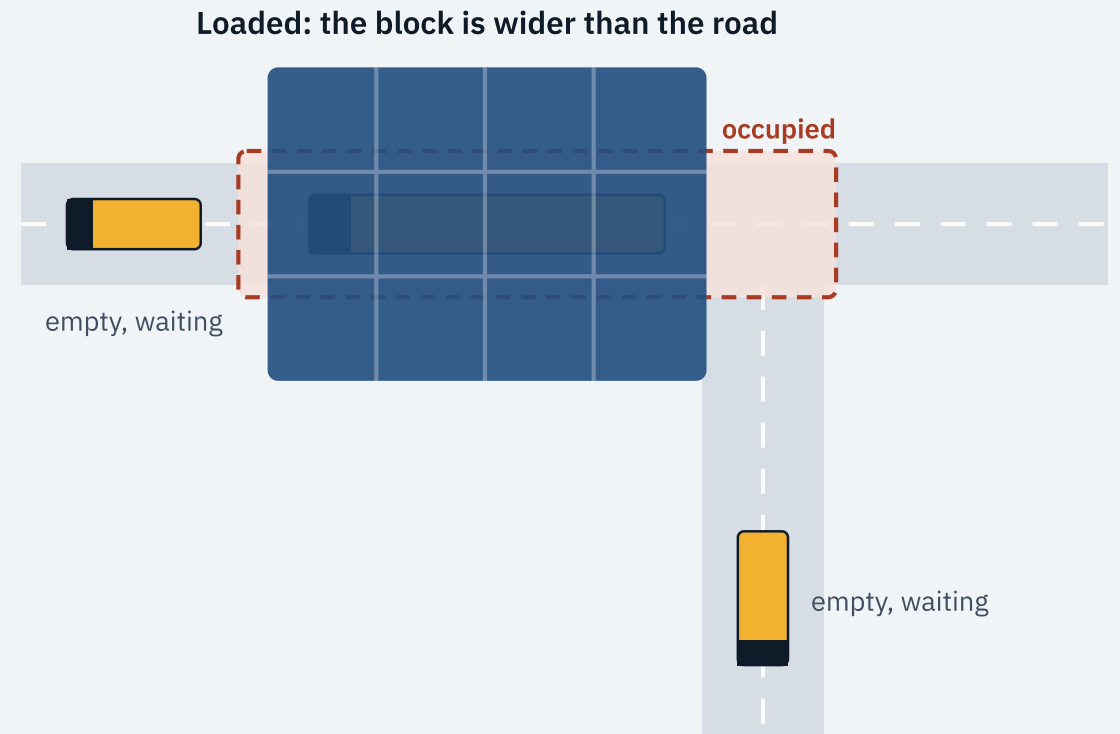
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Direction set 8 Oct 2026 · manifesto v1 frozen 9 Oct 2026 · six sub-studies

Free-flow model: vehicles never meet

Occupancy model: loaded vehicles own the road



The whole thesis in one minute

1

The premise

Shipyards block-transport scheduling almost always assumes a free-flow network: travel times from a distance matrix, vehicles that never interact.

2

The question

When loaded vehicles own the road, what sets capacity, and how should fleet, orchestration and network be designed together?

3

First results

A rule-independent network bound $T1'$, verified on 17,760 points in two yards. Okpo's capacity interval is $[627, 1,720]$: the gap is orchestration. The network-bound regime starts at $K^* \approx 50$ (Okpo) and ≈ 25 (Yantai).

4

Next

Whole-yard task volume (300–600 a day) and the work-zone chart; physical rules and a segmented dock road; the route-free $T1''$; the first road–vehicle exchange-rate table; the first paper.

Capacity is an interval: the network sets its upper edge, orchestration its lower edge.

The order of this briefing (click to jump)

1

Background and premise

Blocks wider than roads; free flow by default; a gap named by peers p. 4

2

Question, claims, hypotheses

Manifesto v1: core question, C1–C5, H1–H7 p. 7

3

Model and theorems

Blocking-time model; theorem list and status p. 11

4

Research map

Six sub-studies, each with its own briefing p. 13

5

Results so far

Bound, capacity interval, regime, dock mouth, retrofits p. 14

6

Papers and venues

Seven chapters, four papers, several candidate venues each p. 16

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Progress and plan

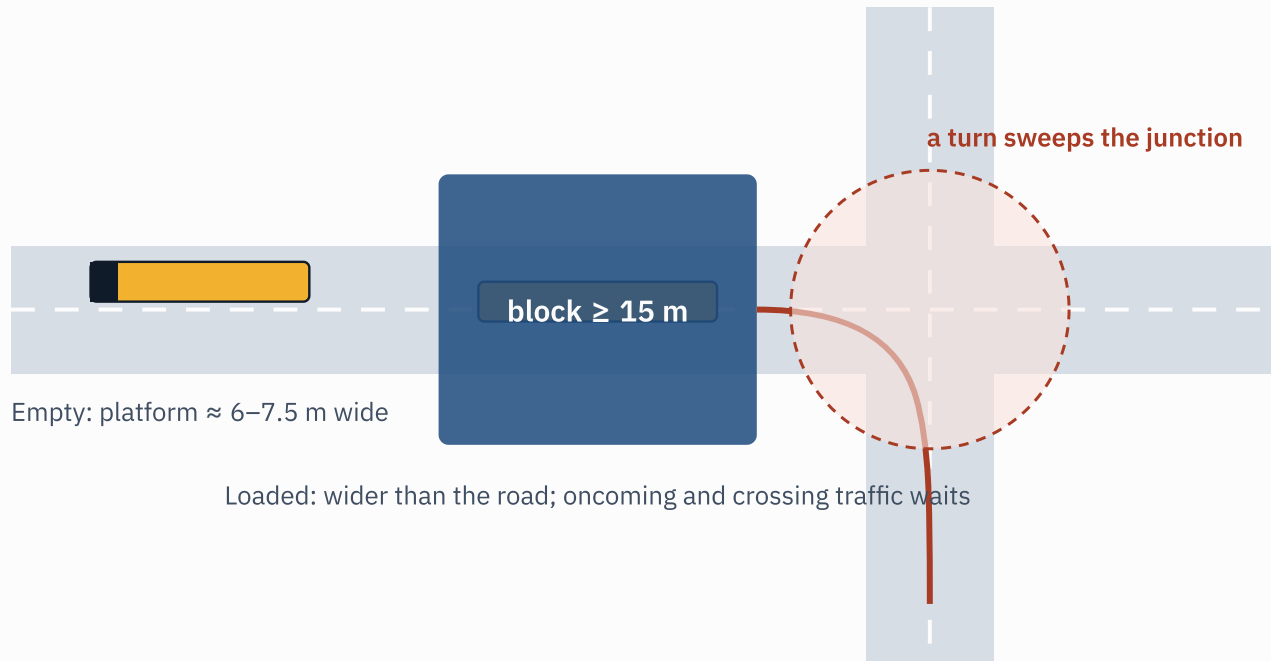
Tasks, eight-week plan, objections, risks and mitigations p. 19

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Earlier work

How fleet sizing and segmentation serve the thesis p. 24

Blocks are far wider than their carriers: once loaded, a transporter holds the whole segment and junction



- **Empty platforms** are about 6–7.5 m wide; one Hanwha Ocean transporter is 21 m × 7.5 m and drives with signallers ahead and behind
- **Loaded outlines** are often above 15 m, e.g. blocks of about 15 × 15 × 12 m
- **The largest carriers** have platforms of about 30 m × 10 m and occupy the road even when empty

The gap between loaded and empty, and between large and small carriers, is particular to shipyards and weak for AGVs and warehouse robots.

The field's default: a free-flow network in which vehicles never interact

TYPE	WORK	TREATMENT	DISTANCE TO OCCUPANCY
Excluded	Meng et al. 2018	Path overlap in the stockyard only; road interference explicitly excluded	free flow
Free flow	Liu et al. 2022; Zhou et al. 2025; Han et al. 2024	Distance matrices or departure and arrival events only; some recommend fleet sizes on that basis	free flow
Turning delay	Moon et al. 2017	Per-vehicle penalties for turning at junctions, changing each vehicle's shortest path	not occupancy between vehicles
Soft penalty	Roh & Cha 2011; Heo et al. 2013	Interference counted in the objective; penalty when two blocks share a segment at the same time	not a structural capacity constraint; loaded and empty not distinguished; no fleet or network design
Single-yard simulation	Tao et al. 2017	Road capacity and “road interference” in a simulation; utilisation rises after reconfiguring vehicles	mechanism unclear; one yard

No one has treated the exclusive use of segments and junctions by loaded vehicles as a structural capacity constraint, or used it to size fleets or upgrade networks (within our search).

Sources: direction document §1.1, checked against the originals on 9 Oct 2026. Full texts: Meng 2018 (line 302), Liu 2022, Zhou 2025 (assumes “traffic congestion not considered”), Roh & Cha 2011 (eqs. 2, 7), Heo 2013, Moon 2017; Han 2024 is a two-page poster, Tao 2017 abstract only. Search scope: two searches of 8 Oct 2026 and the literature library (shipyard scheduling, multi-agent path finding, railway capacity, queueing networks with blocking).

Peers have named the gap, and the yard data show it

A stated gap: Sun, Liang, Chang (2024), *PLOS ONE*

“how to address the interference limitations of roads in the yard remains a limitation of this study. In actual transportation operations, some roads can only accommodate one transporter at a time.”

Chang Daofang's group, Shanghai Maritime University; they may well follow up, so speed matters.

Yard evidence: Wang, Yin, Khan (2024), *JMSE*

288 delays at a Shanghai yard in 2022: transporters take several lanes and other vehicles must give way; the paper assigns each road-environment event (other vehicles, obstacles, pedestrians) a delay of about 15 min (an assigned value, not a calibration anchor).

Other fields: more vehicles jam, road changes move capacity

- **Container-terminal AGV simulation:** going from 60 to 80 vehicles lowered hourly throughput (by about 2%) Cheng et al. 2005
- **Terminal traffic network:** saturation near 100 vehicles; without keep-clear hatching at junctions, deadlocks from 30 Zhou et al. 2017
- **Underground mine:** single-width roadways with passing bays, trucks managed on railway block-section logic Eustace et al. 2021

These are single-site simulations or operating observations; the shipyard's loaded-to-empty footprint ratio is the most extreme of them.

The core question: when loaded vehicles own the road, what sets capacity?

When loaded vehicles occupy the road exclusively, what determines the capacity of a shipyard block-transport system, and how should fleet, orchestration and network be designed together?

The usual view

- Transport capacity = fleet capacity
- Scheduling means assigning jobs to vehicles
- More vehicles are better, with diminishing returns at worst

The road-occupancy view

- Loaded vehicles hold segments and junctions while passing; loaded and empty occupancy are asymmetric
- Capacity is an interval: the network sets the upper edge, orchestration the lower
- Where the network binds, dispatching becomes road scheduling; where the bound is tight, road changes can replace vehicles

What sets capacity?

bounds and interval · ch. 3



How to use it fully?

deadlock-free orchestration · ch. 4



How to expand it?

vehicles or roads · ch. 5



How far does it carry?

cross-yard prediction · ch. 6

Five central claims in one figure

C1 Two upper bounds and the capacity interval

Throughput $\leq \min\{\text{fleet bound } K \cdot H / \bar{c}, \text{network bound}\}$. The network bound follows from the blocking-time structure and is computable by linear programming. Attainable capacity lies in the interval [best deadlock-free rule, network bound]

C2 Fleet-throughput trapezoid

Throughput first rises linearly with fleet size, then reaches a network-set plateau; when waiting occupies roads it declines past a jamming threshold. Free-flow models see neither of the last two

C3 Dispatching becomes road scheduling

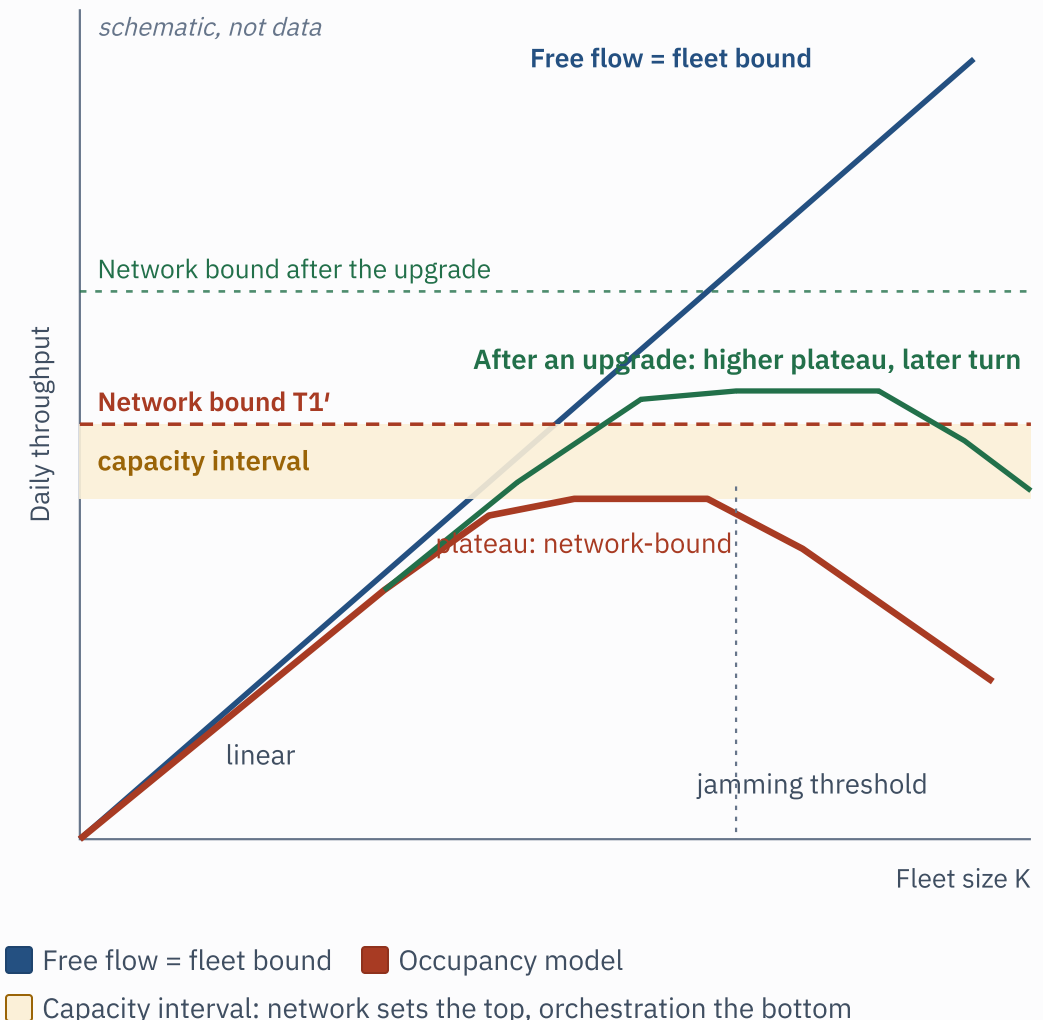
Where the network binds, how much of the bound is attained is decided by traffic orchestration: release order, waiting points, reservation mode, block granularity. Naive dispatching deadlocks

C4 Road-vehicle exchange rate

Where the bound is tight, upgrading key elements can replace vehicles, at the dual price converted into vehicles. When the plateau sits far below the bound, change orchestration first. This is the boundary between C3 and C4

C5 Transferable

The network bound computed straight from the map, and its binding elements, predict plateau, jamming threshold and upgrade value across yards



Seven hypotheses, each conclusive either way (1): H1-H3

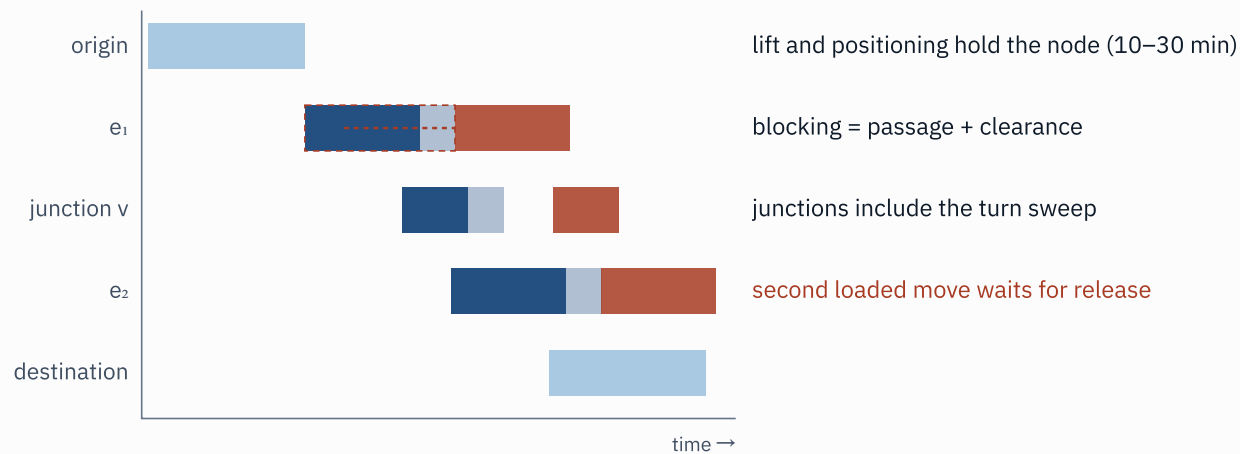
#	HYPOTHESIS	CRITERION	HOLDS / FAILS
H1	Network-bound regime: at realistic daily volumes and fleet sizes, the system is network-bound, at least at erection peaks or near the dock mouth	$K^*(20\%)$: marginal gain first and permanently below 20% of free flow; vehicles needed $\geq K^*$ means network-bound; plus “daily volume / network bound”; whole-yard volumes, one conversion for both yards	Network-bound capacity exists / a density criterion for where free flow suffices
H2	When the decline appears: with off-road waiting and deadlock-free rules throughput never falls (a plateau); when extra vehicles occupy network resources (waiting on roads, incl. for cranes; finite holding area; physical or no deadlock clearing) it falls beyond K_0 , which moves right with holding capacity S ; good orchestration delays K_0	Read saturated curves under no-teleport rules; scan holding capacity S , clearing mode and where vehicles wait for cranes, one at a time	Conditions under which vehicles hurt, and what a holding area is worth in vehicles / a redundancy index (marginal gain vs free flow)
H3	Free-flow error: free flow underestimates completion time and vehicles needed, superlinearly in vehicle density	Service-time gap $\Delta(K) \propto K^\alpha$, $\alpha > 1$; the ratio of vehicles needed read at whole-yard volumes	Systematically wrong at peaks / error under control, with a correction factor

Source: research manifesto v1 §3. H1 is reported in three parts: the capacity interval and its start K^* ; vehicles needed at whole-yard volumes relative to K^* and to free flow; whether the dock mouth caps the yard, which depends on whether vehicles waiting for cranes block the way to other stops.

Seven hypotheses, each conclusive either way (2): H4–H7

#	HYPOTHESIS	CRITERION	HOLDS / FAILS
H4	Road–vehicle exchange rate: where the network binds, one upgrade of a key element is worth at least one vehicle, and dual prices identify the element. Options: segmented blocking, dock-front holding bays, passing bays, one-way loops, widening, a second entrance	Simulated change in vehicles needed vs. the dual-price prediction; agreement is expected only where plateau / bound is near 1	The exchange rate holds / the bottleneck is orchestration; chapter 4 is strengthened
H5	Safe orchestration: naive dispatching deadlocks or nearly so at peaks; a provably safe policy loses little throughput	Deadlocks counted under no-teleport rules; safety cost = loss of the best safe rule relative to the network bound	Safe orchestration is cheap / a cost formula for safety and its conditions
H6	Cross-yard prediction: the network bound computed from the map, with its binding elements, predicts plateau, K^* and upgrade value across yards; topology explains why the bound binds where it does	Leave-one-out; classes by key element: dock-front single lane, door type ...	A two-second LP screens a new yard / what transfers is the bound tool and the key-element classes
H7	Vehicle type and occupancy: large units occupy loaded and empty; small ones yield when empty but run extra empty trips to team up. Case: multi-vehicle moves of giant blocks to floating docks	Compare the best fleet mix across degrees of network constraint	The best fleet mix shifts with network constraint / vehicle type and network can be decided separately

The model: segments and junctions are exclusive resources held for a blocking time



- **Road graph** $G = (V, E)$: segments e and junctions v are exclusive resources
- **A loaded move** holds each resource on its path for a blocking time, as in railway signalling
- **Empty runs by class**: large units occupy even when empty; small ones can pass on passable links
- **No road widths measured**: passability classes (loaded-exclusive / passable by empties / two-way) are scenario parameters; results are functions of the passable share
- **Resource granularity**: long links can be split at their stops with direction locks; the granularity follows the imagery check of the binding elements

One move holds: $\tau_r = \text{pass}_r + \text{clear}_r, r \in \text{path}$

■ loaded move A ■ clearance ■ loaded move B ■ lift, positioning

The theorem list and where each stands

NO.	STATEMENT	STATUS
T1'	The network part of the two bounds: an occupancy-rate LP with width bin-packing rows, crane rows and window-edge terms. Holds for rules satisfying (A1)–(A3); loaded routes fixed	first draft; author rewrites
T1''	Route-free bound: loads also as multicommodity flow; holds for any routing rule	T23 running
T2	Trapezoid law: on corridors and loops, piecewise throughput in K and the jamming threshold under asymmetric loaded/empty occupancy	to do
T3	Unbounded free-flow error: a constructive proof	author
T4	Non-monotonicity: a single-lane corridor with a holding area of capacity S ; under greedy release throughput strictly falls for $K > K_0(S)$	to do
T5	Safe orchestration: reserving to the next safe holding point cannot deadlock, with a loss bound relative to the upper bound	to do (ch. 4)
T6	Road–vehicle exchange rate from dual prices; only upgrades on tight constraints pay. New links use T1''	author
Complexity	Feasibility with deadlines and exclusive resources is NP-hard (reduction from blocking job shop)	to do
Candidates	Dock-mouth proposition: with nominal load $\rho < 1$, blocking-induced starvation destabilises the system; off-road waiting or a segmented dock road restores stability. The other is attainability of T1'	after T22

Source: research manifesto v1 §4 (statements and status). Highlighted: the theoretical core of the first paper. Proofs and text are written by the author.

Research map: one thesis, six sub-studies, each with its own briefing

CH. 2–3

Occupancy model and network bounds

With loads owning the road, what is the rule-free capacity bound?

C1 · T1' · T1''

T1' verified on two yards; T1'' and system T1' under way (T23)

[Open →](#)

CH. 3

Capacity interval and the network-bound regime

Which interval holds attainable capacity, and when does the network bind?

C1 · C2 · H1 · H3

Flag experiment on two yards done; whole-yard volumes and work-zone chart (T22)

[Open →](#)

CH. 3

The dock mouth at erection peaks

Where do vehicles wait at erection peaks, and can the dock mouth hold up the yard?

H1 · H2 · dock proposition

Mechanism established; redrawn with physical rules and a segmented dock road (T22)

[Open →](#)

CH. 4

Traffic orchestration

How to order passages without deadlock and push the lower edge up?

C3 · H5 · T5

Planned; the safe release rule segment_safe comes with T22

[Open →](#)

CH. 5

Fleet-network co-design

Buy vehicles or change roads? How many vehicles is an upgrade worth?

C4 · H4 · H7 · T6

Probe prototype; the first road-vehicle exchange-rate table (T25)

[Open →](#)

CH. 6

Testing across shipyards

Does the bound computed from a map predict across yards?

C5 · H6

Two yards done; group A (six yards) being traced, 15 in all

[Open →](#)

Ch. 2 defines the object



Ch. 3 where capacity comes from



Ch. 4 using it fully



Ch. 5 spending to expand it



Ch. 6 which yards it reaches

Sources: direction document §4 (chapter plan); research manifesto v1; task board T17–T26 (9 Oct 2026). Click a card to open its sub-study briefing.

Results so far: the upper edge stands; the lower edge and the dock mouth are being re-measured under physical rules

17,760 / 0

The bound holds

None of 17,760 occupancy points in two yards exceeds T1' (max 0.973). Width bin-packing rows tighten Okpo's bound by 14% (1,997 → 1,720).

[627, 1,720]

Capacity is an interval

Okpo main scenario: the reservation plateau reaches only 36% of the bound; three road changes move it by +0.2% to +2.4%. The rule, not the road, limits throughput.

$K^* \approx 50 / 25$

The regime starts where the layout says

About 50 vehicles at Okpo, 25 at Yantai. At whole-yard volumes (300–600 a day) vehicles needed may reach K^* ; Yantai's volume is 2–4% of its bound, so free flow suffices: the yards fall on either side of H1.

where to wait

The dock-mouth mechanism

Waiting off-road keeps crane queues local; waiting on the dock road spreads them yard-wide. The “steep drop” so far appears only with teleport clearing and the road as one resource; segmenting the dock road at its stops weakens it sharply, so segmentation becomes Okpo's main scenario.

+10% vs +4%

Control granularity as hidden capacity

When the dock mouth dominates, segmenting the dock road lifts reservation throughput by 10%, widening by only 4%, and passing bays add nothing beyond segmentation (3 seeds; T25 confirms with 10 seeds and the full K grid).

2 types

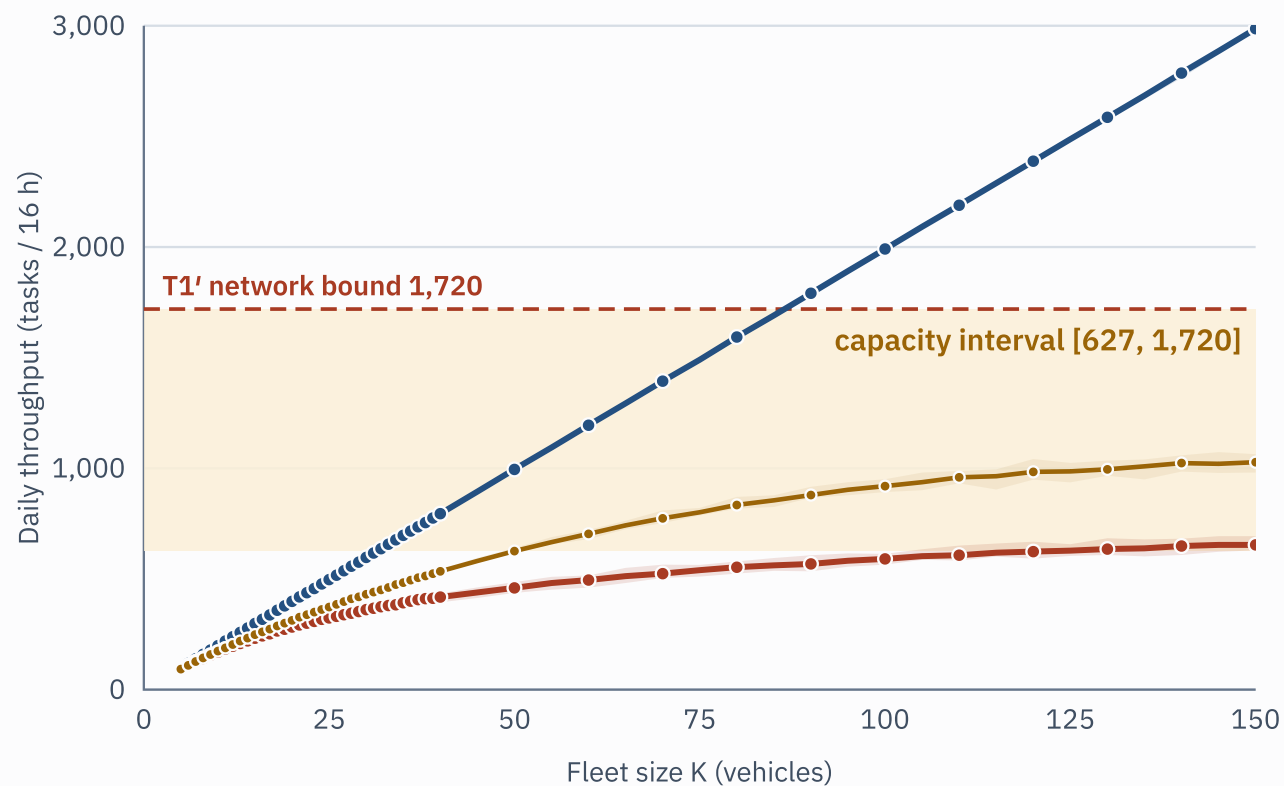
Key-element types emerge

Okpo is a “dock-front single lane” yard (road 161), Yantai a “door” yard (building 014's doors). In both, the author's imagery check of the binding element changed the model, so the check is now a fixed step of chapter 6.

The signature figure: free flow keeps rising, occupancy capacity stays inside the interval

Okpo

Yantai



627–1,720

Capacity interval (a day): lower edge the whole-route reservation plateau (mean of $K = 100\text{--}150$), upper edge $T1'$

36%

The lower edge reaches only this share of the upper: orchestration decides the rest, which chapter 4 aims to win back

$K \approx 50$

Network-bound from here (marginal ratio first below 20%; range 50–60)

Thesis plan: seven chapters and four journal papers answering one question

CH.	TITLE	QUESTION	OUTCOME	CLAIMS · HYP.	PAPER
1	Introduction	Where free flow comes from and why it fails in yards	A literature map across four fields	—	—
2	Occupancy model	How to represent wider loads, lifting at nodes, swept junctions	Blocking-time graph; complexity; public-data instance generator	C1	Paper 1
3	Capacity	Where capacity comes from, when the network binds, when the dock mouth destabilises	T1', T1'', T2–T4; capacity interval and work-zone chart	C1 · C2 · H1–H3	
4	Orchestration	How to assign jobs and order passages without deadlock	Exact model, large-scale search, T5, rolling online	C3 · H5	Paper 2
5	Fleet–network co-design	Buy vehicles or fix roads; large or small vehicles	T6; exchange-rate table; control granularity; bilevel algorithm	C4 · H4 · H7	Paper 3
6	Many yards	Do the laws generalise and predict?	15 public layouts; key-element types; leave-one-out prediction	C5 · H6	Paper 4
7	Conclusion	—	—	—	—

Sources: direction document §4 (chapter plan); research manifesto v1 (claims and hypotheses); the author's review and guidance §4.2–4.3 (outputs of chapters 3 and 6). Highlighted: the theoretical core of the first paper.

Candidate venues for the four papers: the results will decide where each goes

PAPER	FIRST CHOICE	WITH MORE THEORY · ALTERNATIVES	APPLIED	WHAT DECIDES IT
Paper 1 Ch. 2–3 · bounds and capacity interval	<i>TR-E · IISE T</i>	<i>TR-B · TS</i> <i>TR-C · EJOR</i>	<i>OE</i>	Whether attainability of T1' and the dock-mouth proposition are proved: if so, TR-B / TS
Paper 2 Ch. 4 · orchestration	<i>TR-C · TS</i>	<i>T-ASE · EJOR</i> <i>C&OR</i>	<i>T-ITS · OE</i>	T5's loss bound and the algorithm's scale: strong theory, TS; strong algorithms, TR-C / T-ASE
Paper 3 Ch. 5 · fleet–network co-design	<i>TR-E · IISE T</i>	<i>TR-B · EJOR</i> <i>Omega</i>	<i>C&IE</i>	Whether the T6 exchange rate becomes a theorem; whether the bilevel algorithm carries optimality gaps
Paper 4 Ch. 6 · many yards	<i>OE · C&IE</i>	<i>IJPR · FSMJ</i>	<i>JSPD · IJNAOE · JMSE</i>	Number of yards and predictive accuracy: 15+ yards with leave-one-out prediction holding, IJPR tier

Sources: the author's review and guidance §4.2 (paper 1's venues and conditions); direction document §4 and §8.2; the rest is a list of options for the author. TR-B/C/E = Transportation Research Part B/C/E; TS = Transportation Science; IISE T = IISE Transactions; EJOR = European J. Operational Research; C&OR = Computers & Operations Research; T-ASE / T-ITS = IEEE Trans. Automation Science and Engineering / Intelligent Transportation Systems; OE = Ocean Engineering; C&IE = Computers & Industrial Engineering; IJPR = Int. J. Production Research; FSMJ = Flexible Services and Manufacturing J.; JSPD = J. Ship Production and Design; IJNAOE = Int. J. Naval Architecture and Ocean Engineering; JMSE = J. Marine Science and Engineering.

Algorithms, and what the thesis brings back to three research communities

Algorithms (chapters 4–5)

- **Exact model:** CP-SAT interval variables, one no-overlap constraint per segment and junction; ground truth on small instances
- **Large scale:** integrated assignment-and-sequencing neighbourhood search; priority-based planning for conflicts; compared with “schedule in free flow, then repair conflicts”
- **Safety layer:** the T5 reservation policy as a shell around any heuristic
- **Rolling online:** re-plan under uncertain completion times; release with forecasts
- **Co-design:** outer level picks upgrades and fleet, inner level evaluates by orchestration; T6 dual prices rank the outer options
- **Optional frontier:** LLM-evolved orchestration rules (EoH, ReEvo, FunSearch; no shipyard use found)

Framing: “we extend X to Y and find that Z changes under Y”

Multi-agent path finding

A new agent class: footprint multiplies when loaded, long lifts at endpoints, fleet size to be chosen; existing large agents have fixed shapes and no tasks

Railway capacity

Blocking times and compression carried to trackless road networks with asymmetric loaded and empty occupancy

Queueing networks with blocking

A real system that breaks the vehicle–hole symmetry, because empty and loaded vehicles occupy differently

Progress: two days of flag experiments, one full review, a frozen manifesto, and three corrections running in parallel

WHEN	ITEM	OUTPUT	STATUS
10-08	T17–T18 flag experiment (Okpo); T1 tightened to T1' (width bin-packing, crane rows)	Fleet-size–throughput curve; T1' draft proof	done
10-09	T19–T21 second yard, Yantai; erection at crane cadence; in- and out-of-span erection and ρ curves	Two-yard comparison; dock-mouth mechanism	done
10-09	Author's full review; manifesto v1 frozen; imagery check of road 161 and building 014	Three first-order corrections; segmented dock road as main scenario	done
T22	Safe release rule segment_safe; segmented dock road; whole-yard volumes (300 / 450 / 600, peak $\times 2$)	Work-zone chart; capacity interval; dock mouth redrawn	running
T23	System T1' (no sampling noise); route-free T1'' and the value of route choice	Bound figures; top-10 dual prices	running
T24	Literature anchors: Yim 2008, Shen 2018; Tao 2017 full text compared item by item	External comparison table; introduction	running
T25	Retrofit simulations (segmenting, holding bays, passing bays, widening); door granularity; three-resource dock network	First exchange-rate table	queued
T26	Okpo rev 1833 (floating docks, branches); floating-dock scenario; finite holding area S	Final Okpo map; $K_0(S)$ for H2	queued

Sources: task board T17–T26 and its “awaiting the author” section (9 Oct 2026); reviews T17–T21; research manifesto v1 (thesis-core tag manifesto-v1).

Eight-week plan: two weeks of corrections, first paper submitted within 3–4 months

WHEN	THE AUTHOR	COMPUTATION	OUTPUT (PLACE IN THE THESIS)
Weeks 1–2	Manifesto ✓; imagery check of road 161 and building 014 ✓; start annotating group A	T22 physical rules, segmenting, whole-yard volumes; T23 system T1' and T1"; T24 literature anchors	Ch. 3 work-zone chart; dock-mouth section; bound figures
Weeks 3–4	Rewrite the T1' proof; if T22 supports it, write the dock-mouth proposition (three-resource network)	T25 retrofit simulations; T26 Okpo rev 1833 and floating docks	First exchange-rate table; ch. 5 prototype; final Okpo map
Weeks 5–8	Write the first paper	External comparison table; T1' and plateaus for the first group-A yards	Paper 1 submission draft; first ch. 6 data

The first paper's four results: (1) model and T1'; (2) capacity interval and work zones; (3) the dock mouth: where vehicles wait and how the dock road is segmented; (4) the rule separating C3 from C4, and the first exchange-rate table.

Sources: the author's review and guidance §4.2 (first paper) and §4.4 (first two weeks and eight weeks); task board T22–T26. “Submit within 3–4 months”: direction document §12.

The likeliest objections, and the answers

OBJECTION	ANSWER
Yard conflicts are done already Roh & Cha 2011; Heo 2013; Moon 2017; Tao 2017	They treat crossings as soft penalties in path search or turn time for one vehicle, or observe “road interference” in one yard's simulation; none writes exclusivity as a structural capacity constraint, gives bounds, compares rules or handles deadlock, or designs fleet and network.
MAPF already has large agents Li et al. 2019	Their shapes are fixed and they carry no pickup-and-delivery tasks; none combines a load-dependent footprint, long lifts and fleet sizing. The thesis uses them as a source to build on.
Yards have too few vehicles to jam	That is what H1 tests. At whole-yard volumes (300–600 a day), Okpo's vehicles needed may reach or pass $K^* \approx 50$; Yantai's volume is 2–4% of its bound. The yards falling on either side is itself the result; T22 is recomputing on this basis.
The dock-mouth drop is a modelling artifact	So far the drop appears only with teleport clearing and the dock road as one resource, and we say so. T22 redraws it with a non-teleporting safe rule and segmented blocking; the result is stated as a conditional reading of where vehicles wait and how the road is segmented, and as a design lever.
No measured data	The theorems need no data; task volumes use published whole-yard figures (Yim 2008; Shen 2018); vehicle counts are compared with yards after normalising by cycle time; many public layouts; the passable share is scanned over its whole range.
Exclusivity is too strong	Exclusivity is a parameter: results are functions of the passable share; resource granularity (whole segment, split at stops, doors) is a scenario parameter too, chosen by the imagery check.

Sources: direction document §10; the author's review and guidance §2.1–2.3, §5 and appendix B; Tao 2017 read at abstract level (T24 fetches the full text).

Risks and mitigations

RISK	MITIGATION
At whole-yard volumes the occupancy model needs far more vehicles than yards run (about 30)	Normalise by cycle time first; if a gap remains, report both readings: calibrate the model (segmenting, shifts, width distribution), or treat it as chapter 4's orchestration target. Either way, “is free flow enough” gets a real magnitude.
Under physical rules and segmenting, Okpo's dock-mouth drop weakens or vanishes	State the result as a conditional reading of where vehicles wait and how roads are segmented, plus a design lever, not a single yard's threshold; group-A yards with narrower, longer dock roads add cases.
Competition: Tao, Jiang et al. 2017 simulated “road interference” and adjusted the transporter fleet	T24 fetches the full text; the introduction separates item by item: formalised exclusive resources and blocking times, rule comparison and deadlock, bounds, multiple yards.
The Chang Daofang group (Sun 2024) follows up	No 2025–2026 follow-up found; the first paper goes out within 3–4 months as planned, with a preprint.
Only two yards so far	Compute T1' for the six group-A yards first (fast), simulations follow; the imagery check of key elements is a fixed step for every yard.

Sources: the author's review and guidance §5 (risks and mitigations); task board, “author's imagery check” (key-element review as a fixed step of chapter 6). “About 30” is the reported scale of yard fleets, checked in T24.

The results ladder: every chapter's floor is a paper on its own; the ceiling aims at top journals

	FLOOR (DONE ONCE DONE)	TARGET	CEILING
3 Capacity	Two bounds and the trapezoid on one yard two yards in hand	T1', T1'', T2–T4 proved; capacity interval and work-zone criterion	Attainability of T1' and the dock-mouth proposition
4 Orchestration	Exact model and large-scale search beating “free-flow schedule, then repair”	T5 safe policy with small loss; the lower edge pushed to yard per-vehicle productivity	LLM-evolved rules beating hand-written ones and transferring across yards
5 Co-design	Grid evaluation of fleets and upgrades	T6 exchange rate and key elements (incl. control granularity)	Bilevel co-design with optimality gaps
6 Many yards	8 yards	15+ yards; key-element types and leave-one-out prediction	A simple formula for jamming thresholds from topology

Sources: direction document §11; the author's review and guidance §4.2–4.3.

Earlier work serves as tools; the thesis starts from a new premise

ASSET	USE IN THE THESIS
Fleet sizing (MY-B)	ALNS and CP-SAT code start chapter 4; price data feed chapter 5's vehicle types and costs
Coupled-carriage ideas	Chapter 5, H7: large carriers against teams under occupancy
Centreline networks and the tracing tool	Production tool for chapter 6's layouts
Remote-sensing segmentation (MY-A, published)	Optional: faster layout extraction for many yards

Open the fleet-sizing (MY-B) briefing →

Open the segmentation (MY-A) briefing →

In a shipyard, a loaded transporter does not drive on the road; it owns the road. Capacity is capped by the network, not the fleet.

Thank you.

Source: direction document §9 (use of earlier assets). The thesis starts from the premise of §1; earlier work is used as tools and cited.

References (1/3)

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